FIITJEE Students From All Programs have bagged 33 in Top 100, 77 in Top 200 and 205 in Top 500 All India Ranks. FIITJEE Performance in JEE (Advanced), 2015:
4551 FIITJEE Students from All Programs have qualified in JEE (Advanced), 2015.

ALL INDIA INTEGRATED TEST SERIES

FIITJEE **JEE (Advanced), 2016**

PART TEST – III

Paper 1

Time Allotted: 3 Hours **Maximum Marks: 180**

- Please read the instructions carefully. You are allotted 5 minutes specifically for this purpose.
- You are not allowed to leave the Examination Hall before the end of the test.

INSTRUCTIONS

A. General Instructions

1. Attempt ALL the questions. Answers have to be marked on the OMR sheets.
2. This question paper contains Three Parts.
3. **Part-I** is Physics, **Part-II** is Chemistry and **Part-III** is Mathematics.
4. Each part is further divided into two sections: **Section-A & Section-C**
5. Rough spaces are provided for rough work inside the question paper. No additional sheets will be provided for rough work.
6. Blank Papers, clip boards, log tables, slide rule, calculator, cellular phones, pagers and electronic devices, in any form, are not allowed.

B. Filling of OMR Sheet

1. Ensure matching of OMR sheet with the Question paper before you start marking your answers on OMR sheet.
2. On the OMR sheet, darken the appropriate bubble with black pen for each character of your Enrolment No. and write your Name, Test Centre and other details at the designated places.
3. OMR sheet contains alphabets, numerals & special characters for marking answers.

C. Marking Scheme For All Three Parts.

(i) **Section-A (01 – 10)** contains 10 multiple choice questions which have one or more than one correct answer. Each question carries **+3 marks** for correct answer. There is no negative marking.

(ii) **Section-C (01 – 10)** contains 10 Numerical based questions with answers as numerical value from **0 to 9** and each question carries **+3 marks** for correct answer. There is no negative marking.

Name of the Candidate	
Enrolment No.	

Useful Data

PHYSICS

Acceleration due to gravity	$g = 10\ m/s^2$
Planck constant	$h = 6.6 \times 10^{-34}$ J-s
Charge of electron	$e = 1.6 \times 10^{-19}$ C
Mass of electron	$m_e = 9.1 \times 10^{-31}$ kg
Permittivity of free space	$\varepsilon_0 = 8.85 \times 10^{-12}\ C^2/N\text{-}m^2$
Density of water	$\rho_{water} = 10^3\ kg/m^3$
Atmospheric pressure	$P_a = 10^5\ N/m^2$
Gas constant	$R = 8.314\ J\ K^{-1}\ mol^{-1}$

CHEMISTRY

Gas Constant	R	=	$8.314\ J\ K^{-1}\ mol^{-1}$
		=	0.0821 Lit atm $K^{-1}\ mol^{-1}$
		=	$1.987 \approx 2$ Cal $K^{-1}\ mol^{-1}$
Avogadro's Number	N_a	=	6.023×10^{23}
Planck's constant	h	=	6.625×10^{-34} J·s
		=	6.625×10^{-27} erg·s
1 Faraday		=	96500 coulomb
1 calorie		=	4.2 joule
1 amu		=	1.66×10^{-27} kg
1 eV		=	1.6×10^{-19} J

Atomic No: H=1, He = 2, Li=3, Be=4, B=5, C=6, N=7, O=8, N=9, Na=11, Mg=12, Si=14, Al=13, P=15, S=16, Cl=17, Ar=18, K =19, Ca=20, Cr=24, Mn=25, Fe=26, Co=27, Ni=28, Cu = 29, Zn=30, As=33, Br=35, Ag=47, Sn=50, I=53, Xe=54, Ba=56, Pb=82, U=92.

Atomic masses: H=1, He=4, Li=7, Be=9, B=11, C=12, N=14, O=16, F=19, Na=23, Mg=24, Al = 27, Si=28, P=31, S=32, Cl=35.5, K=39, Ca=40, Cr=52, Mn=55, Fe=56, Co=59, Ni=58.7, Cu=63.5, Zn=65.4, As=75, Br=80, Ag=108, Sn=118.7, I=127, Xe=131, Ba=137, Pb=207, U=238.

Physics

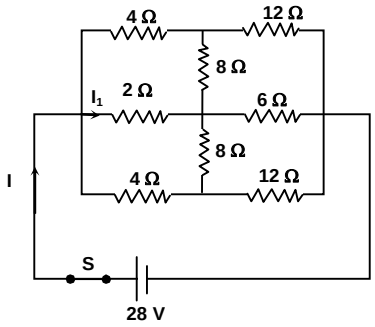
PART – I

SECTION – A

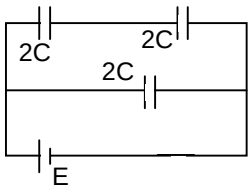
Straight Objective Type

This section contains 10 multiple choice questions numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which ***ONE or MORE*** are correct.

1. In the circuit shown, the current
(A) I = 8 A
(B) I = 7 A
(C) I_1 = 3.5 A
(D) I_1 = 4 A

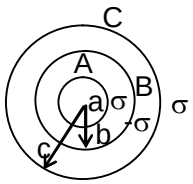


2. Three capacitors are connected with the source of electromotive force E as shown in the figure. Heat Loss is



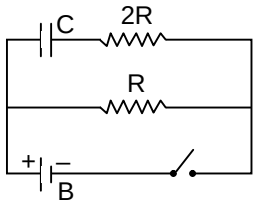
(A) $\frac{1}{2}CE^2$ (B) $\frac{3}{2}CE^2$
(C) $3CE^2$ (D) $2CE^2$

3. The adjacent figure shows charged spherical shells A, B and C having charge densities σ,-σ, σ and radii a, b, c respectively. If $V_A=V_c$ then c equal to (assuming a = 10 cm, b=20 cm,)
(A) 30 cm (B) 50 cm
(C) 25 cm (D) 35 cm



Space for rough work

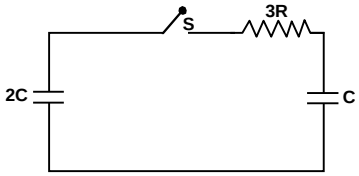
4. In the circuit shown, when the switch is closed, the capacitor charges with a time constant



(A) RC (B) 2RC

(C) $\frac{2RC}{3}$ (D) $\frac{8RC}{3}$

5. The capacitor 2C is charged to potential E_o while the capacitor C is uncharged at t = 0. At this instant, the switch S is closed.



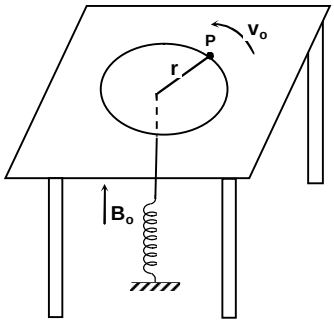
(A) Charge on the capacitor C is $\frac{2}{3}CE_0$ in steady state

(B) Charge on the capacitor 2C is $\frac{4}{3}CE_0$ in steady state

(C) Charge on the capacitor C is $\frac{1}{3}CE_0$ in steady state

(D) Charge on the capacitor 2C is $\frac{5}{3}CE_0$ in steady state

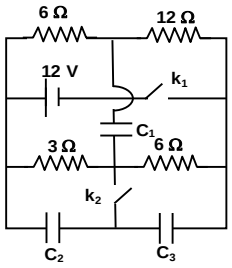
6. A particle mass m charge q is moving on a circular path on the surface of a frictionless table with speed v_o where a magnetic field B_o exists uniformly over the whole region. It is attached by a string which passes through a hole in the table to a spring as shown. The spring is stretched by x_o. If now the magnetic field is increased slowly to $2B_o$,



(A) The extension in the spring will increase
(B) The speed of the particle will increase
(C) The speed of the particle will decrease
(D) The kinetic energy of the particle will decrease

Space for rough work

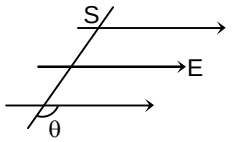
7. Given $C_1 = 2\mu F$; $C_2 = 4\mu F$, $C_3 = 8\mu F$. C_1, C_2, C_3 are initially uncharged. Initially all the keys are open. Then



(A) In steady state, after closing k_1 alone the charge in capacitor C_1 is zero.
(B) In steady state, after closing k_1 alone the charge in capacitor C_2 and C_3 are $8\mu C$ and $4\mu C$.
(C) In steady state, after closing k_1 alone the current drawn from the cell is 2A.
(D) In steady state, after closing k_1 and k_2 the current drawn from the cell is 1A.

8. A proton is fired form origin with velocity $\vec{v} = v_0\hat{j} + v_0\hat{k}$ in a uniform magnetic field $\vec{B} = B_0\hat{j}$. In the subsequent motion of the proton
(A) its z coordinate can never be negative
(B) its x coordinate can never be positive
(C) its x and z coordinates cannot be zero at the same time
(D) its y coordinate will be proportional to its time of flight.

9. A plane surface of area S is inclined at an angle θ with a uniform field $\vec{E}$ as shown in the figure. Find the magnitude of flux of $\vec{E}$ over S.



(A) ES cos θ (B) ES sin θ
(C) – ES cos θ (D) – ES sin θ

10. The magnetic field intensity inside a long solenoid is 1 T. If the current per turn of the winding is 1 A, Find the number of turns in one meter.

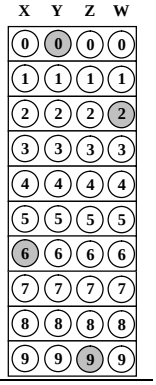
(A) $\frac{1}{4\pi}.10^7$ (B) $\frac{1}{4\pi}.10^6$
(C) $\frac{1}{2\pi}.10^6$ (D) $\frac{1}{8\pi}.10^7$

Space for rough work

SECTION – C

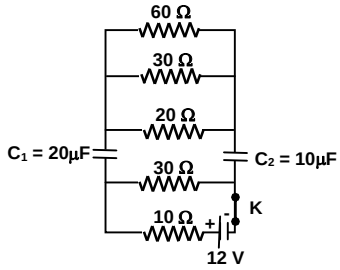
Integer Answer Type

This section contains 10 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown



1. A charged particle is projected in a magnetic field $\vec{B} = \left(x\hat{i} + 4\hat{j}\right)10^{-2}\,T$. The acceleration of the particle is found to be a $\vec{a} = \left(\frac{8}{3}\hat{i} - 2\hat{j}\right) m/s^2$. Find the value of x.

2. For the circuit arrangement shown in the figure, find the potential difference (in V) across C_2 in the steady state condition.

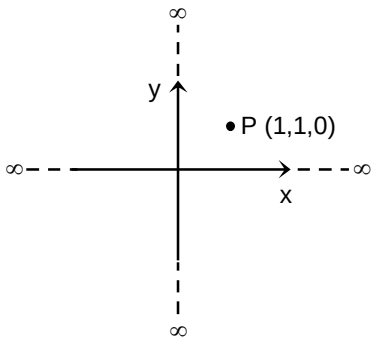


3. A steady current I goes through a wire loop PQR having shape of a right angle triangle with PQ= 3x, PR= 4x and QR= 5x. If the magnitude of the magnetic moment is $k(Ix^2)$. Find the value of k.

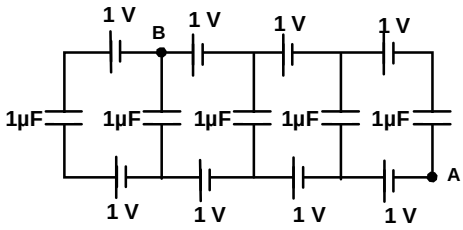
4. In a moving coil galvanometer, torque on the coil can be expressed as $\tau \propto i$, where i is current through the wire. The rectangular coil of the galvanometer having number of turns N, area A and moment of inertia I is placed in magnetic field B. The torsion constant of spring in the galvanometer is C and charge Q is passed through the coil almost instantaneously. If the maximum angle through which coil deflected is $\frac{KNABQ}{\sqrt{4CI}}$. (Ignore the damping in mechanical oscillations).

Space for rough work

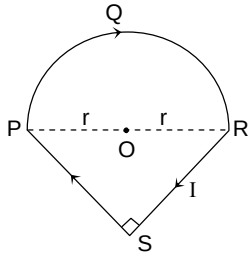
5. There are two infinitely long & broad sheets, one is lying on xz plane and another one is lying on yz plane. In both the sheets, current is moving in positive z direction having linear density (current per unit perpendicular length) equal to $\frac{\sqrt{2}}{\pi}\times 10^7$ Ampere/meter. Find the magnitude of magnetic field (in Tesla) at point P shown in the figure.



6. The potential difference $V_B - V_A$ for the circuit shown in the figure is $\frac{24}{k}$ V. Then find the value of k.

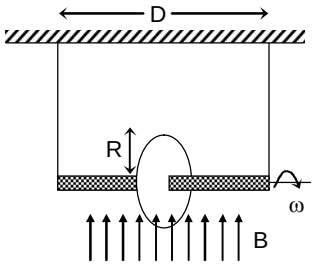


7. A steady current I = 10 A goes through a wire loop PQRS. Part PQR is semi-circle of radius r = 1 m. RS = SP and ∠RSP = 90°. Find the magnetic field at O in 10^{-6}T to the nearest integer.

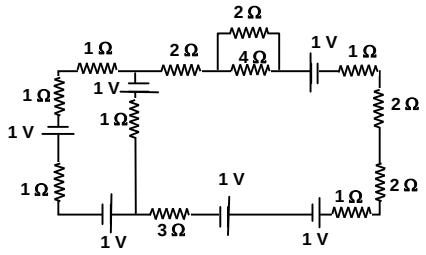


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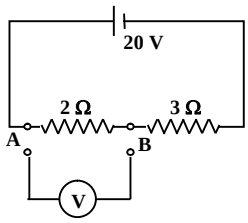
8 The axle of a circular wheel of radius R is held horizontally by two identical strings of equal lengths magnetic by a distance D. The tension in each string is T_0. The rim of the wheel carries a total charge + Q distributed uniformly on it. The wheel is vertical and is kept in a uniform vertical magnetic field $\vec{B}$. It is now rotated at an angular speed ω. If the string break at a tension of $3T_0/2$, If the maximum possible value of ω at which the wheel can be rotated without breaking a string is $\frac{KT_0D}{QR^2B}$. Then find the value of K.



9. Calculate the ratio of magnitude of current in 3Ω resistance is to magnitude of current in 4Ω resistance



10. When a voltmeter, as shown in n the figure is connected across a 2Ω resistance, it reads 5 volts. Find the internal resistance of the voltmeter (in ohm).



Space for rough work

Chemistry

PART – II

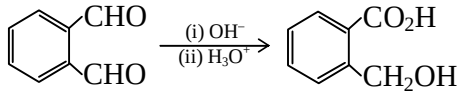
SECTION – A

Straight Objective Type

This section contains 10 multiple choice questions numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which ***ONE or MORE*** are correct.

1. Select the correct statement(s), among the following given statements
   (A) $CH_3CH_2CH_2I$ will react more readily than $(CH_3)_2CHI$ for SN^2 reactions
   (B) $CH_3CH_2CH_2Cl$ will react more readily than $CH_3CH_2CH_2Br$ for SN^2 reactions
   (C) $CH_3CH_2CH_2CH_2Br$ will react more readily than $(CH_3)_3C - CH_2Br$ for SN^2 reactions
   (D) $CH_3 - O - C_6H_5 - Br$ will react more readily than $NO_2 - C_6H_5 - CH_2Br$ for SN^2 reaction

2. Consider following reaction



   This:
   (A) is a crossed Cannizzaro reaction.
   (B) is intramolecular Cannizzaro reaction.
   (C) causes substitution of H^- by OH^- in one–CHO group and is thus oxidised while second –CHO undergoes an addition of H^- and is thus reduced to alcohol.
   (D) resembles aldol type condensation reaction.

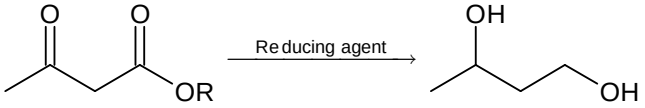
3. Lucas test is used to make distinction between 1°, 2° and 3° alcohols:

$$ROH + HCl(conc.) \xrightarrow{\text{anhy. } ZnCl_2} \underset{(\text{White turbidity})}{RCl\downarrow} + H_2O$$

   This show that
   (A) ROH behaves as a base
   (B) Greater the value of pK_a(alcohol), greater the reactivity with conc. HCl & thus sooner the formation of white turbidity
   (C) Alcohol which reacts fastest with Na metal, will give turbidity at fastest rate
   (D) Alcohol which gives red colour during victor mayor test will give turbidity at slower rate in Lucas test than those giving blue or white colour during victor mayor test

Space for rough work

4. Reducing agent(s) which can be used to carry out following conversion is/are

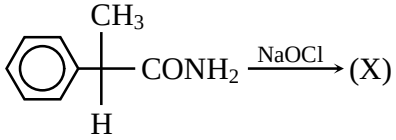


(A) $NaBH_4$ (B) $LiAlH_4$
(C) H_2/Ni (D) B_2H_6/THF

5. Which of the following statement(s) about D(+) glucose is/are true?
(A) Naturally occurring glucose is dextrorotatory (B) It reduces ammoniacal $AgNO_3$ solution
(C) On polymerisation it forms galactose (D) It forms cyanohydrin on reaction with HCN

6. Which of the following statements is/are correct?
(A) In S_N2 mechanism, the rate determining step involves two species
(B) The S_N2 mechanism involves the formation of intermediate carbocation
(C) The S_N2 mechanism is accompanied with Walden inversion
(D) The S_N2 mechanisms may involve intramolecular rearrangement

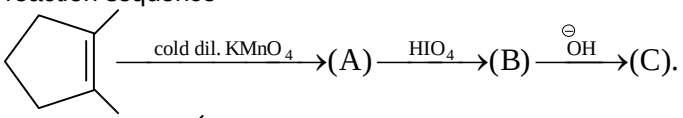
7. Which of the following statements are true about the major product (X) in the reaction?

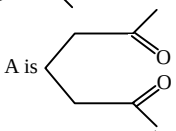


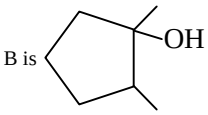
(A) (X) is a cyclic amide
(B) (X) has an asymmetric centre
(C) (X) is a primary amine
(D) (X) can also be obtained by treatment of NH_3 with $C_6H_5CH(CH_3)Br$

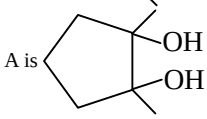
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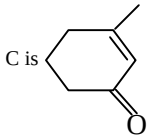
8. In a reaction sequence



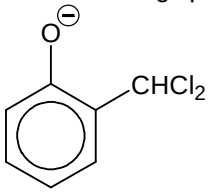
(A) 

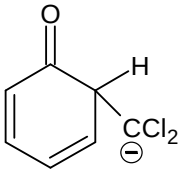
(B) 

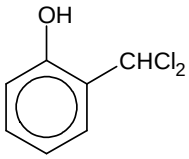
(C) 

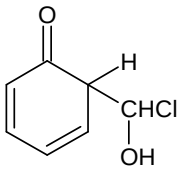
(D) 

9. When phenol is reacted with $CHCl_3$ and NaOH followed by acidification, salicylaldehyde is obtained. Which of the following species is/are involved in the above mentioned reaction as intermediates?

(A) 

(B) 

(C) 

(D) 

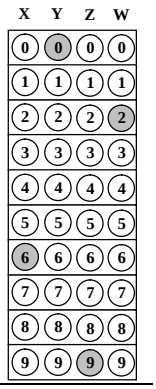
10. Which of the following statement (s) is/are correct?
(A) The solubility of n – alcohol in water decreases with an increase in molecular weight
(B) The solubility of n – alcohol in water increases with an increase in molecule weight
(C) The reactive proportion of the hydrocarbon part in alcohols increases with increasing molecular weight which permits enhanced hydrogen bonding with water
(D) The reactive proportion of the hydrocarbon part in alcohols decreases with increasing molecular weight which reduces hydrogen bonding with water

Space for rough work

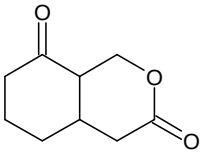
SECTION – C

Integer Answer Type

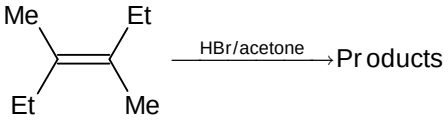
This section contains 10 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown



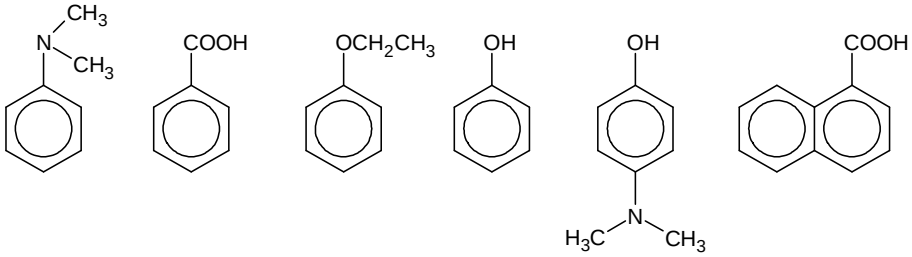
1. How many molecules of CH_3MgX are consumed in the reaction with one molecule of the following, in dry ether?



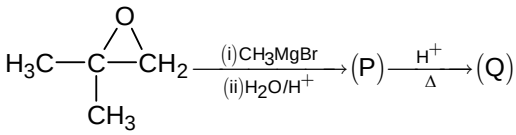
2. How many products will be formed in the following reaction?



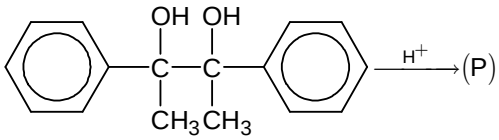
3. Amongst the following the total number of compounds soluble in aqueous NaOH is



Space for rough work

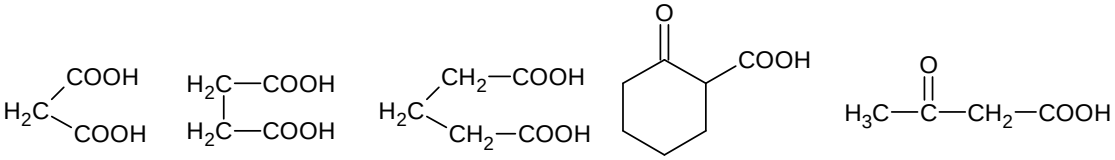
4. 

How many hyperconjugating structures of "Q" are possible?

5. 

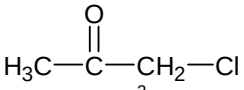
How many α - H to carbonyl groups are there in the product "P"?

6. How many of the following acids easily undergo decarboxylation on heating?

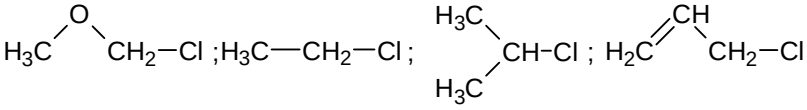


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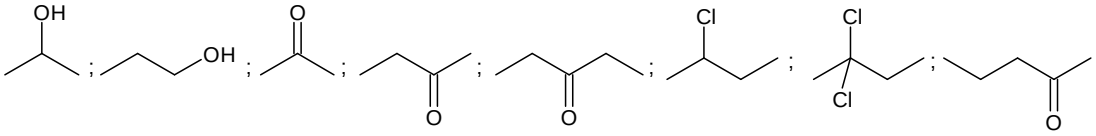
7. How many of the following compounds are less reactive than



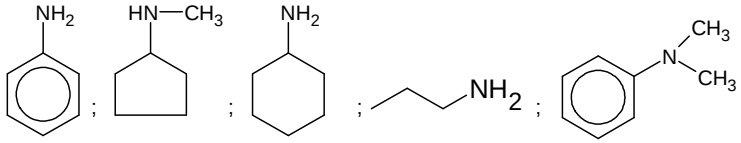
towards SN^2 reaction:



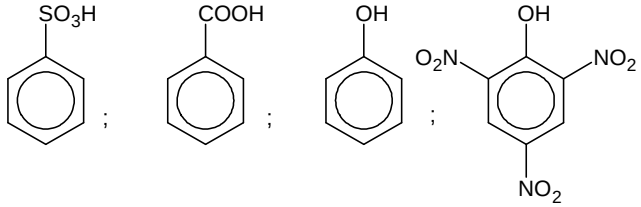
8. How many of the following compounds give positive haloform test?



9. How many of the following give positive carbyl amine test?



10. How many of the following acid liberate CO_2 from $NaHCO_3$ solution?



Space for rough work

Mathematics

PART – III

SECTION – A

Straight Objective Type

This section contains 10 multiple choice questions numbered 1 to 10. Each question has 4 choices (A), (B), (C) and (D), out of which ***ONE or MORE*** are correct.

1. Which of the following pair of function is identical?

(A) $f(x) = \cos(2\tan^{-1}x)$; $g(x) = \frac{1-x^2}{1+x^2}$

(B) $f(x) = \frac{2x}{1+x^2}$; $g(x) = \sin(2\cot^{-1}x)$

(C) $f(x) = \tan x + \cot x$; $g(x) = 2\,\text{cosec}\,2x$

(D) $f(x) = e^{\ln\left(\text{sgn}\cot^{-1}x\right)}$; $g(x) = e^{\ln\left[1+\{x\}\right]}$

(where {.}, [.] denotes fractional part function and greatest integer function)

2. Let $f(x) = (x-1)^4(x-2)^n$; $n \in N$ f(x) is

(A) local minima at x = 2; if n is even
(B) local minima at x = 1; if n is odd
(C) local maxima at x = 1 if n is odd
(D) local minima at x = 1 if n is even.

3. $f(x) = 2x^3 - 3(a+1)x^2 + 6ax - 12$ has local maximum at x_1 and local minimum at x_2 and if $2x_1 = x_2$ then value of a is

(A) 1
(B) $\frac{1}{2}$
(C) – 1
(D) 2.

4. Which of the following functions has/have removable discontinuity at x = 1

(A) $f(x) = \frac{1}{\ln|x|}$
(B) $f(x) = \frac{x^2-1}{x^3-1}$
(C) $f(x) = 2^{-2^{\frac{1}{1-x}}}$
(D) $f(x) = \frac{\sqrt{x+1}-\sqrt{2x}}{x^2-x}$

Space for rough work

5. $f(x)=\frac{[x]+1}{\{x\}+1}$ for $f:\left(0,\frac{5}{2}\right]\to\left(\frac{1}{2},3\right]$; where [.] represents G.I.F. and {.} $\to$ fractional part of x

(A) f(x) is injective discontinuous function
(B) f(x) is surjective non differentiable function
(C) $\min\left(\lim_{x\to1^-}f(x),\lim_{x\to1^+}f(x)\right)=f(1)$
(D) max (x values of point of discontinuity) = f(1)

6. $f(x)=\left(\frac{x}{2+x}\right)^{2x}$, then

(A) $\lim_{x\to0}f(x)=-4$ (B) $\lim_{x\to\infty}f(x)=2$
(C) $\lim_{x\to\infty}f(x)=e^{-4}$ (D) $\lim_{x\to1}f(x)=\frac{1}{9}$.

7. If $y=10^{10^x}$ and $\frac{1}{y}\frac{dy}{dx}=10^x\cdot\lambda$, then λ is

(A) ln 10 (B) $(\ln 10)^2$
(C) $e^{\ln(\ln 10)^2}$ (D) $(\log_{10}e)^2$.

8. For function $f(x)=\frac{\ln x}{x}$ which of the following statement are correct

(A) f(x) has horizontal tangent at x = e (B) f(x) cuts the x axis only one point
(C) f(x) is many one function (D) f(x) has one vertical tangent.

9. If $F(x)=\frac{1}{x^2}\int_4^{x^2}\left(4t^2-2F'(t)\right)dt$ then F′(4) equals to

(A) $\frac{32}{9}$ (B) $\frac{64}{9}$
(C) $\frac{F(8)}{28}$ (D) $\frac{11}{28}f(8)$

Space for rough work

10. $f(x)=\begin{cases} x+\frac{1}{2}; & x<0 \\ 2x+\frac{1}{3}; & x>0 \end{cases}$

[.] denotes greatest integer function

(A) $\lim\limits_{x\to 0}\left[f(x)\right]=0$

(B) $\lim\limits_{x\to 0} f(x)$ does not exist

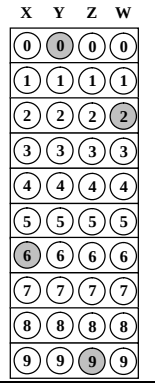
(C) $\left|\lim\limits_{x\to 0} f(x)\right|$ exist

(D) $\lim\limits_{x\to 0}\frac{\left[f(x)\right]}{x}$ does not exist.

SECTION – C

Integer Answer Type

This section contains 10 questions. The answer to each of the questions is a single-digit integer, ranging from 0 to 9. The appropriate bubbles below the respective question numbers in the ORS have to be darkened. For example, if the correct answers to question numbers X, Y, Z and W (say) are 6, 0, 9 and 2, respectively, then the correct darkening of bubbles will look like the as shown



1. If $x^2 + y^2 = 1$ and maximum value of $x + y$ is $\frac{\sqrt{2}}{3}\lambda$ then value of λ is _____

2. $f\left(x+\frac{1}{2}\right)+f\left(x-\frac{1}{2}\right)=f(x)\ \forall\ x\in R$ then the period of f(x) is _____

3. Number of points where the function f(x) = max (| tanx |, cos |x|) is non differentiable in the interval $(-\pi, \pi)$ is _____

4. Tangent of acute angle between the curves $y = |x^2 - 1|$ and $y = |x^2 - 3|$ at their point of intersection is $\frac{4\sqrt{2}}{k}$ then value of k is ____

Space for rough work

5. $\int \frac{\cos^4 x\, dx}{\sin^3 x\left(\sin^5 x+\cos^5 x\right)^{\frac{3}{5}}} = -\frac{1}{2}\left(1+\cot^A x\right)^B + c$ then AB is equal to _____

6. $\int_{-\pi}^{\pi}(\cos px - \sin qx)^2\, dx$ is equal to $m\pi$ then value of m _____

7. If $S_n = \frac{1}{2n} + \frac{1}{\sqrt{4n^2-1}} + \frac{1}{\sqrt{4n^2-4}} + \ldots \frac{1}{\sqrt{3n^2+2n-1}}$ is $n \in N$ then $\lim_{n\to\infty} S_n$ is $\frac{\pi}{n}$ then N is _____

8. $\int_{-1}^{1} \sin^3 x \cos^2 x\, dx$ is equal _____

9. The lateral edge of a regular hexagonal pyramid is 1 cm if the volume is maximum then the height must be $\frac{M}{\sqrt{3}}$; find the value of M _____

10. If $F''(x) = -f(x)$ and $g(x) = f'(x)$ and $F(x) = \left(f\left(\frac{x}{2}\right)\right)^2 + \left(g\left(\frac{x}{2}\right)\right)^2$ and given $F(5) = 5$ then $F(10)$ is equal to

Space for rough work